

Augmented Reality applications for Warehouse Logistics

Bengt Mueck¹, Matthias Höwer¹, Werner Franke^{1,2}, Wilhelm Dangelmaier¹

{mueck|qwert}@hni.upb.de, werner.franke@siemens.com, whd@hni.upb.de

¹Heinz Nixdorf Institute, Fuerstenallee 11, D-33102 Paderborn, Germany

²Siemens AG, Logistics & Assembly Systems, Postbox 4549, D-90451 Nürnberg, Germany

Abstract

The costs for material flow and logistics generate a major part of the prime costs for a product. Thereby warehouse and distribution logistics take an outstanding position in inner- and outer operational supply chain. Under the circumstances that on average 2/3rd of the operational costs for logistics are accounted for warehouse management tasks, it is clear evidently that these costs have a decisive influence on the competitive position of a company. As part of a cooperative project of the “Logistics & Assembly Systems” of Siemens AG, one of the leading companies in logistics and production automation, and the department for Computer Integrated Manufacturing of the University of Paderborn, the worlds first Augmented Reality based picking system was developed in the labs at Heinz Nixdorf Institute Paderborn. The prototype that emanated from the development process has been approved in extensive tests under laboratory conditions. The evaluation of the test results verified the potential to improve the work flow and picking efficiency with an Augmented Reality based picking system.

1 Motivation

The majority of the costs for logistics are determined by a human failure coefficient. Value creating processes in warehouses are often accomplished by unskilled workers, which for example are hired from a temporary employment company. It is advantageous to hire time-workers for warehouse management tasks, like picking, because these processes are very dependent on the company's order situation. On the other hand there are many problems which are directly hinged on the assignment of temporary employees. The picking rate, i.e. the number of items picked within a certain time, is below the picking rate for skilled labours. Furthermore the number of aborted picks rises, which cause a better part of the costs for returns and customer service.

Also the increasing competition, for example via the currently discussed enlargement of the European Union with states from the former Eastern Bloc, yields to additional requirements to optimize the structures in manufacturing companies. Production lines and warehouse storage systems are automated to the greatest

possible extend. Anyhow there are key processes in the supply chain that are only can be handled only by humans. Be it the described picking process, inventory, system maintenance or other similar tasks.

On that point new technologies like Augmented Reality can be extremely useful as a support system. The chances to minimize the human errors within these processes by the use of Augmented Reality are tremendous.

2 State Of The Art

Augmented Reality (AR) is a new technology that provides the potential to add virtual information into the employees working space, with the help of a Head Mounted Display (HMD). The AR system used for the development of this project is the ARToolKit framework [2]. The basic AR technology is fairly simple. The employees view is recorded with a mini camera attached to the HMD. The images are transferred to a wearable PC that is running the AR system. The AR system searches the image for so called AR markers, identifies the markers position and draws the additional information perspectively correct onto the image. The output picture in the HMD shows then the newly created image. Figure 1 shows a basic illustration of the AR concept.

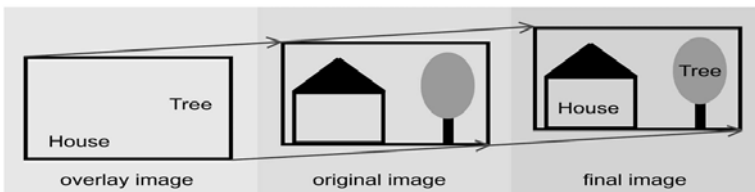


Figure 1: Illustration of the AR Concept

ARToolKit is an application that has been developed at the University of Washington since the late 90's. There are a number of research projects that use the ARToolKit technology. These projects reach from ARToolKit based tracking [5], to more consumer oriented projects like the BlackMagicBook [1], a virtual book that uses 3D models which can be explored with the help of a HMD.

A major part of the research that was done in this project was focused on the optimization opportunities within Warehouse Management processes with the help of the described AR technology. All Warehouse Management support systems that can be found on today's market follow the same goal. Error ratios that are the consequence of human failings, should be minimized to the least possible amount. The predominant portion of processes in a Warehouse Management System is already fully automated [3]. Figure 2 shows an overview of Warehouse Management processes. The product storage and stock removal for example is completely optimized with digital control units, which are provided by a number of worldwide operating companies. These units range from shelf access equipment or fully automatic small part stocks, to driverless transport systems within the storage facility [4].

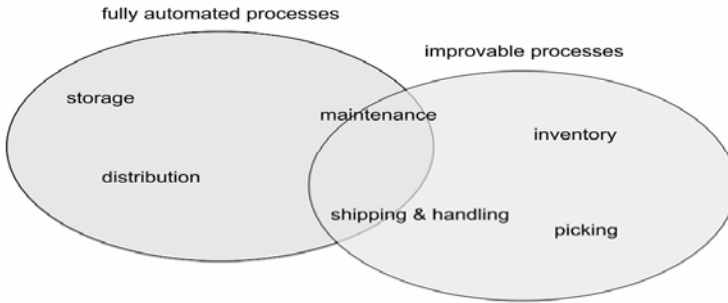


Figure 2: Overview of classic warehouse management processes

Other processes, like picking or order combination, aren't completely optimized so far. Support systems like „Pick By Voice“ (PBV) or „Pick By Light“ (PBL) are two methods that try to minimize the existing problems [Figure 3]. PBV is a support system that guides the employee acoustically, step by step, through the entire picking process. The major advantage of PBV is that there is no curtailing for the employee and he can use both hands during the picking process.



Figure 3: „Pick By Light“ (left), „Pick By Voice“ (right); Source: [4]

Contrary to that PBL is a visual support system. Rack compartments are modified with digital counters which show the amount that has to be picked from the compartment. The major problem with PBL is that it is not unlimited possible to use the PBL system simultaneously with multiple workers. So it mostly works for serial picking processes [4].

3 Scenario Selection

All storage processes that have been researched for optimization within this project have a sizable part of manual tasks. The following processes have been researched for further benchmarking:

1. Stock Receipt: The products are delivered palletized and labelled with AR-Toolkit markers. The employee gets the order confirmation visualized in a

HMD and verifies the pallet content of delivered products with the help of the ARToolKit marker.

2. **Picking:** Each rack compartment is labeled with an ARToolkit marker. The picking employee receives the customer order within a HMD. Further on a warehouse overview (Floorplan) and a detailed article description, including a picture (Description), are visualized within the HMD. Right when a product is localized within the rack, the employee gets a positive feedback from the Augmented Reality system by the recognition of the ARToolkit markers.
3. **Shipping and Handling:** Each package that has to be shipped gets labelled with an ARToolkit marker. The customers order is digitally visualized by the Augmented Reality support system for the employee. Besides the packing list, the order is virtually visualized in a schematical illustration in the HMD. The employee then compares the virtual order with the existing package.
4. **Inventory:** Each rack compartment is labelled with an ARToolkit marker. The employee for inventory receives the scheduled inventory product list within a HMD. Further on a warehouse overview (Floorplan) and a detailed article description, including a picture (Description), are visualized within the HMD. Right when a product is localized within the rack, the employee gets a positive feedback from the Augmented Reality system by the recognition of the ARToolkit markers.

All four scenarios have been elaborated step by step and were evaluated under criteria which are momentous for the use in Warehouse Management Systems (Table 1).

Criteria: / Scenario:	Stock Receipt	Picking	S & H	Inventory
Conditions within the storage facility (Light, Space offering)	+	o	+	O
Increase of efficiency	o	++	+	++
Assistance from the use of AR	o	+	o	+
Feasibility in ARToolkit	o	++	+	++
Endangering from the use of AR hardware	+	o	++	-
Negative external influences	--	++	++	O

Table 1: Evaluation of support scenarios *

The results of this research clearly pointed out that the best setting for an AR based support system would be the development of a picking system implemented under ARToolkit and since the inventory processes are very similar to picking, the developed system can easily be adjusted to fulfil the requirements for an AR based inventory system.

* Evaluation method: -- very bad; - bad; o average; + good; ++ very good

4 Concept „Virtual Pick By Light“

The main goal during the development of the „Virtual Pick by Light“-application (VPBL) was to provide the picking comfort of the PBV systems on the one hand and combine that with the visual support of the existing PBL systems on the other hand. Based on this purpose four support features have been developed for implementation:

1. **Picklist:** The Picklist is the digitized version of the customers order. It contains only the rack compartment number as well as the the amount that has to be picked for this product. The product which has to be currently picked by the employee is clearly highlighted from the other ones within the list.
2. **Description:** The Description is visualized dependent on the active picklist position from the current order. Besides the rack compartment number it includes a short description and a picture of the product.
3. **Floorplan:** The Floorplan as well is generated dependent on the picklist position. The employee gets a visualized warehouse overview. The overview indicates both the position of his last pick (red) and the position of the current pick (green).
4. **AR-Visualization:** The AR-Visualization is the last step of verification for the picking process. The ARToolKit marker which labels the rack compartments is used as a final visual feedback to support the correctness of the pick. Once a rack compartment is recognized it is automatically highlighted with a transparent green rectangle, which is visualized and perspectively aligned with the help of ARToolKit.

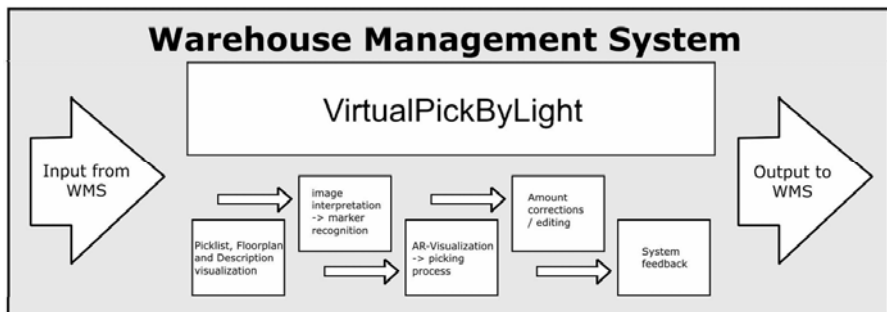


Figure 4: Flow chart VPBL

The four support features provide all functions that are included in the PBV and PBL systems, except the acoustic guidance which has been replaced by a visual floorplan feature. Figure 4 shows an abridgement of the functional coherence of the four support features as well as the informational flow within the Warehouse Management System.

A quite simple formula allows to compare a VPBL with other picking application like PBV or PBL. Picking a customer order in a warehouse is a sequence of work steps accomplished by a picker. Typical steps are “to take an order”, “to find a bin location”, “to identify and check the product” etc. There are types of work steps which will be done only one time for each order $S_{onetime} = \{^o s_1, ^o s_2, \dots, ^o s_h\}$ and other types of steps which will be done multiple times in an order fulfillment $S_{multi} = \{^m s_1, ^m s_2, \dots, ^m s_n\}$. The number of different significant steps is given by $n, h \in \mathbb{N}$. Let $t_{^o s_i} \in \mathbb{R}^+$ be the average execution duration of a step $^o s_i \in S_{onetime}$ and analog $t_{^m s_j} \in \mathbb{R}^+$ be the average execution duration of a single step $^m s_j \in S_{multi}$ which is repeated $r_{^m s_j} \in \mathbb{N}$ times by picking a typical order.

Similar to this it makes sense also to define average error rates $e_{^o s_i} \in \mathbb{R}^+$ for each significant single and $e_{^m s_j} \in \mathbb{R}^+$ for each multiple repeated step. Depending on the specific processing and goods portfolio in each warehouse mostly quality and speed aspects have different importance. This applies to the whole picking sequence and could be furthermore individual for each work step in the sequence. For example to find a bin location in a roomily picking area has different importance to same work step in a comparatively small area. A set of warehouse context specific weight factors for speed $p_{^o s_i}, p_{^m s_j} \in \mathbb{R}^+$ and quality

$q_{^o s_i}, q_{^m s_j} \in \mathbb{R}^+$ can express this different interest of aspects per step. This leads to the following formula for a Quality-Speed-Coefficient (qsC) of a picking principle in a specific warehouse:

$$qsC = \sum_{^o s_i \in S_{onetime}} (t_{^o s_i} \cdot p_{^o s_i} + e_{^o s_i} \cdot q_{^o s_i}) + \sum_{^m s_j \in S_{multi}} r_{^m s_j} (t_{^m s_j} \cdot p_{^m s_j} + e_{^m s_j} \cdot q_{^m s_j})$$

The computing and comparison of qsC for VPBL and other concurrent picking principles can help to find the best fitting solution in a specific warehouse and idiosyncratic requirements for speed and quality.

5 Conversion

The system environment that was used for the conversion of the project is based on a standard notebook, running Microsoft Windows XP. For the visual data input a standard notebook webcam was utilized. The system output occurred over a consumer market HMD. All system input and feedback from the employee was realized with wireless keypad (Figure 5). With the keypad it is possible to enable

and to disable the support features, decrease the amount and navigate through the picklist to correct unintended system feedbacks.

5.1 Implementation

The four main features of the VPBL application that have been described in the concept, are based on the ARToolKit framework. The Picklist, Description and Floorplan functions thereby merely use OpenGL and glut libraries and are permanently visualized for the employee. Opposed to that the AR-Visualization uses the full spectrum of the ARToolKit framework and is not permanently visible during the picking process. Figure 6 shows a screenshot of the running application and the AR-Visualization in the form of a (green) rectangle.

The delineation of the three permanent support features have been implemented optionally and can be displayed via keyboard commands when necessary. The Picklist has been implemented on the upper right side of the screen. The current item is always highlighted from the other items displayed in the Picklist. The employee gets an easy to understand graphical feedback from the system when an item was correctly picked or for example no product was found in the rack compartment. All positive picks are displayed with a green overlay pad in the amount field. Negative or incomplete picks are displayed with red or orange pad.

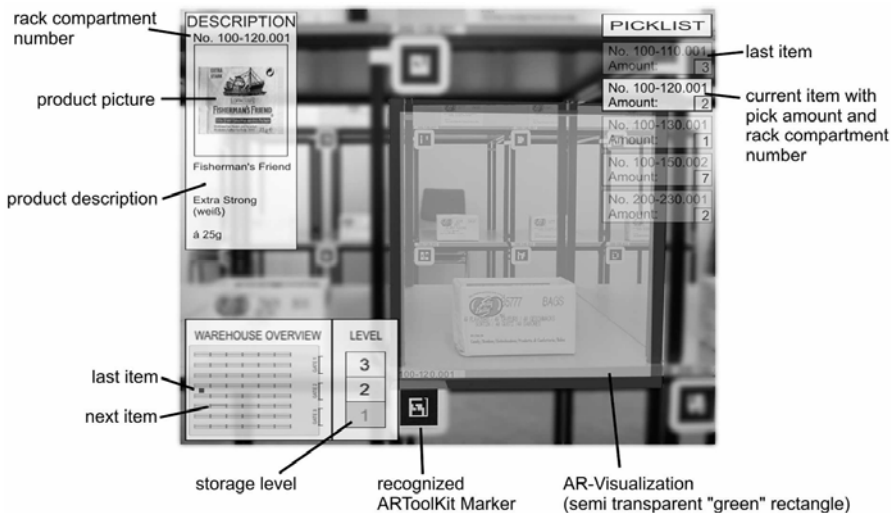


Figure 5: Detailed view of the picker in front of a rack compartment

The Description support feature is drawn in the upper left quadrant of the screen. The semitransparent illustration is dependent from the selected picklist item. The same applies for the floorplan support function. It is shown in the lower left half of the screen. It is also semitransparent and allows the employee to check the last,

next and level position of the rack compartments. Figure 5 shows in detail how the support systems have been implemented for the VPBL application.

5.2 Experiments and Evaluation

The developed prototype has been approved in extended tests under laboratory conditions. The probands had to absolve alternating tests with and without the VPBL system. The test scenario was a provisional warehouse with two shelf rows with 5 columns each row and two rack compartment levels in each column. The hardware that was used for the evaluation matched the setup described in section five and is also shown in Figure 6 (right picture).

Each rack compartment was individually labelled with a storage code. The storage code is build from three different numerical pads. The first three digits, for example “100”, indicate the shelf row. The next three digits, like “130”, identify each column and the last three digits, “001” or “002” guide the employee in which rack compartment level the current compartment is located. To assure a counterbalanced test environment all products are stored in uniform cardboard boxes (Figure 6). Within this test scenario the probands had to pass through six test runs under two different picking conditions. “Case 1” was the developed Augmented Reality picking application. “Case 2” was a standard picking scenario with a printed pick-list. In “Case 2” the probands had then to identify the rack compartment from the storage code, in “Case 1” they were guided by the VPBL system. For all picking runs the time and, if necessary, the number of failed picks has been acquired.

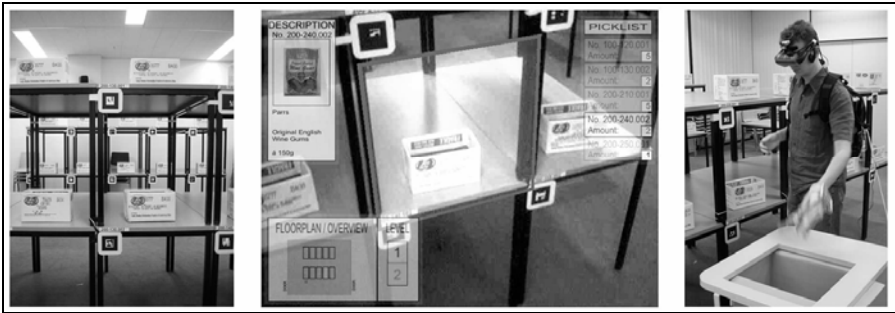


Figure 6: Experiment set-up and accomplishment

From the centralized chronology during the picking process a strong learning curve for each proband is clearly recognisable. Especially promising is the fact that the learning curve is significantly better in the use of the VPBL system. The time improvement with the Augmented Reality system was 39% in average from the first to the second run. From run two to run three the improvement was still recognisable with 12% in average.

Although the absolute figures showed that the standard picking was approximately 20% faster in the test scenario, this wasn't expected otherwise due to the size of the used „warehouse” environment. With a growing complexity of a warehouse assembly and a increasing number of similar product versions, a time based improvement with the use of an Augmented Reality system is the logical consequence.

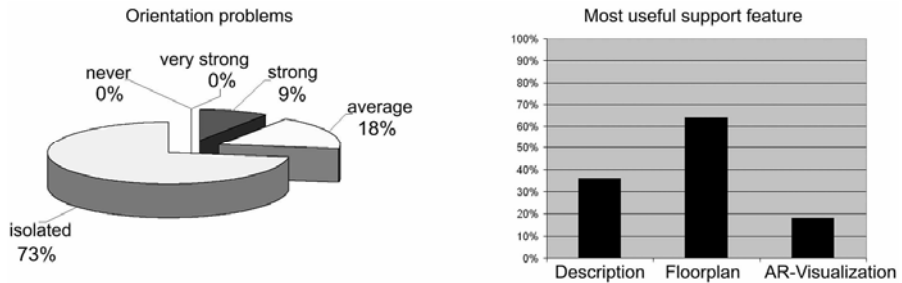


Figure 7: Results

Another very promising result was the decrease of failed picks. Once a wrong product was stored in one of the rack compartments, 0% of fail picks happened with the help of the VPBL application. Without the support system 50% of the wrong products have been picked anyways. This is a positive advice which synergies to improve the efficiency of a picking process can be realized with the use of an Augmented Reality based picking system.

To evaluate the implemented support functions each proband had to answer a detailed questionnaire following to the six test runs. This again proved the positive results of the practical first half of the system evaluation.

Orientation problems were a major concern. But even with the non optimized hardware system orientation problems appeared only isolated by 73% of the test candidates. It is very likely that these were sourced from the consumer HMD that was used for the testing.

Another question focused on the three support systems besides the picklist. More than 60% of the respondents judged the Floorplan as the key feature of the support system, followed by the Description with more than 35% and the AR-Visualization with a little under 20% (multiple answers possible)(Figure 7).

6 Conclusion and Prospects

The applications for an Augmented Reality technology are proven with the implementation of the VPBL picking system. Furthermore the technology can be widened almost unedited to other support applications in Warehouse Management Systems. The shortest path navigation, the Augmented Reality based rack compartment search or the additional storage code and item description information

are also applicable in stock activities for storing, outhousing or storage reorganization. Similar Augmented Reality Systems can be implemented for Stock Receipt, Shipping & Handling, Quality Control or administrative tasks in Warehouse Management Systems.

The result that the warehouse overview (Floorplan) is the most selected feature also shows how important navigation assistance is especially for unskilled and untrained workers.

The Augmented Reality system won't be able to replace manual processes in the added value chain, but it is possible to reduce the human factor to a minimum with the help of visual support system.

There are still a number of features that can be improved and added to the existing prototype. The used hardware will be improved for the use in a Warehouse Management environment. The prototype will be optimized under usability aspects and improved with a number of features that were researched in the testing scenario as a result of the evaluation.

Literature

- [1] HITLabNZ, Human Interface Technology Laboratory New Zealand, University of Canterbury, 2003 [<http://www.hitlabnz.org>]
- [2] hit lab, Human Interface Technology Lab, University of Washington, USA, 2003. [<http://www.hitl.washington.edu>]
- [3] Warehouse Management, M. ten Hompel, T. Schmidt, Springer 2003.
- [4] Siemens AG, Siemens Logistics & Assembly, Offenbach, Germany 2004.
- [5] Technische Universität Wien, Institut für Softwaretechnik und Interaktive Systeme, Wien, Österreich, 2004 [<http://www.ims.tuwien.ac.at>]